

LOAN APPROVAL PREDICTION USING MACHINE LEARNING

Student Details :

Name: Gangamma

Roll Number: 20241CSD0148

Department: B.Tech CSE (Data Science)

College: Presidency University, Bengaluru

Internship Organization: Alfido Tech

Project Title : Loan Approval Prediction Using Machine Learning

Objective :

The objective of this project is to develop a machine learning model that predicts whether a loan application will be approved or rejected based on applicant information. The model helps financial institutions make faster, more accurate, and data-driven lending decisions while minimizing financial risks.

Dataset Description :

The Loan Approval dataset contains demographic, financial, and personal information about loan applicants. The target variable indicates whether the loan application was approved or rejected.

Dataset Features

- Gender
- Marital Status
- Dependents
- Education
- Self Employed
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Amount Term
- Credit History
- Property Area
- Loan Status

Tools and Technologies Used :

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Google Colab / Jupyter Notebook
- GitHub

Data Preprocessing : The dataset was cleaned and prepared before model training.

Steps Performed :

1. Imported the dataset using Pandas.
2. Checked for missing values.
3. Filled missing values using suitable techniques.
4. Removed duplicate records.
5. Converted categorical data into numerical format using Label Encoding.
6. Normalized and standardized numerical data when necessary.
7. Split the dataset into training and testing datasets.

Exploratory Data Analysis (EDA) :

Exploratory Data Analysis was performed to understand the dataset and identify patterns. Analysis Performed

- Distribution of loan approval status.
- Applicant income distribution.
- Loan amount distribution.
- Credit history analysis.
- Relationship between education and loan approval.
- Relationship between income and loan approval.

Observations :

- Applicants with good credit history had higher approval rates.
- Higher income applicants showed better chances of loan approval.
- Loan approval depended on multiple financial factors.

Methodology :

The project followed a structured machine learning workflow.

Step 1: Data Collection

The Loan Approval dataset was collected and loaded into the Python environment.

Step 2: Data Cleaning

Missing values and duplicate records were handled to improve data quality.

Step 3: Data Exploration

Visualizations and statistical summaries were generated to understand data patterns.

Step 4: Feature Engineering

Relevant features were selected and transformed for machine learning.

Step 5: Model Development

Different machine learning algorithms were trained using the prepared dataset.

Step 6: Model Evaluation

Models were evaluated using accuracy score, confusion matrix, precision, recall, and F1-score.

Step 7: Interpretation

Important factors influencing loan approval were identified using feature importance analysis.

Feature Engineering :

Feature engineering was performed to improve prediction performance.

Techniques Used :

- Label Encoding
- Feature Selection
- Data Transformation
- Target Variable Encoding

Important Features :

- Credit History
- Applicant Income
- Loan Amount
- Education
- Property Area

Model Building :The dataset was divided into:

- Training Data (80%)
- Testing Data (20%)

The machine learning models were trained on the training dataset and evaluated on the testing dataset.

Process :

1. Train Model
2. Predict Loan Status
3. Evaluate Performance
4. Compare Results

Algorithms Used :

Logistic Regression:A statistical classification algorithm used for binary prediction.

Decision Tree Classifier : A tree-based algorithm that classifies applicants based on decision rules.

Random Forest Classifier : An ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy.

Among the tested algorithms, Random Forest achieved the best performance.

Model Accuracy :

The trained model achieved an accuracy of approximately 80%–90%, depending on preprocessing and model parameters.

Evaluation Metrics :

- Accuracy Score
- Precision
- Recall
- F1 Score

Confusion Matrix : A confusion matrix was generated to evaluate prediction performance.

Components :

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

Feature Importance Analysis :

Feature importance was analyzed using the Random Forest algorithm.

Top Influencing Features :

1. Credit History
2. Applicant Income
3. Loan Amount
4. Property Area
5. Education

Credit History was identified as the most important feature affecting loan approval.

Key Findings :

- Credit history plays a major role in loan approval decisions.
- Applicants with higher incomes generally have a better chance of approval.
- Loan amount influences approval probability.
- Machine learning models can accurately automate loan approval prediction.
- Random Forest performed better than other classification algorithms.

Recommendations :

Based on the project findings, the following recommendations are suggested:

1. Financial institutions should prioritize credit history during loan evaluation.
2. Applicant income and repayment capacity should be carefully assessed.
3. Machine learning models can be integrated into banking systems for faster processing.
4. Models should be periodically retrained with updated data.
5. Additional financial indicators can be included to improve prediction accuracy.
6. Automated systems should support, not replace, final human decision-making.

Conclusion :

This project successfully developed a machine learning model for predicting loan approval status. Through data preprocessing, feature engineering, and model training, accurate predictions were achieved. The results demonstrated that credit history, applicant income, and loan amount significantly influence loan approval decisions. The project highlights the potential of machine learning in improving efficiency, consistency, and accuracy in financial decision-making.

GitHub Repository Link :

<https://github.com/Gangamma-yn/Task-2-LOAN-APPROVAL-PREDICTION-.git>

Google Colab/Jupyter Notebook Link :

<https://colab.research.google.com/drive/1wOC3ySrrmJLKikZapLc4DEv6rcPv7Lwv?usp=sharing>

```
df=pd.read_csv("loan_prediction.csv")
df.head()
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	1.0	Urban	Y
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	N
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	Y
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	Y
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	Y

Handle caiss imbalance

```
smote=SMOTE(random_state=42)

x_train_smote,y_train_smote=smote.fit_resample(
    x_train_processed,
    y_train
)
print("Before SMOTE:")
print(y_train.value_counts())

print("\nAfter SMOTE:")
print(y_train_smote.value_counts())
```

```
*** Before SMOTE:
Loan_Status
1    337
0    154
Name: count, dtype: int64

After SMOTE:
Loan_Status
1    337
0    337
Name: count, dtype: int64
```

Train Logistic regression

```
model=LogisticRegression()
model.fit(x_train_smote,y_train_smote)
```

LogisticRegression

Parameters

Model Selection

```
# Get class predictions (0 or 1)
y_pred = model.predict(x_test_processed)

# Get probabilities for ROC-AUC
y_prob = model.predict_proba(x_test_processed)[: , 1]

print("\n----- LOGISTIC REGRESSION RESULTS -----")

print("Accuracy Score:")
print(accuracy_score(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

----- LOGISTIC REGRESSION RESULTS -----

Accuracy Score:
0.8211382113821138

Classification Report:

	precision	recall	f1-score	support
0	0.72	0.68	0.70	38
1	0.86	0.88	0.87	85
accuracy			0.82	123
macro avg	0.79	0.78	0.79	123
weighted avg	0.82	0.82	0.82	123

Confusion Matrix:
[[26 12]
[10 75]]

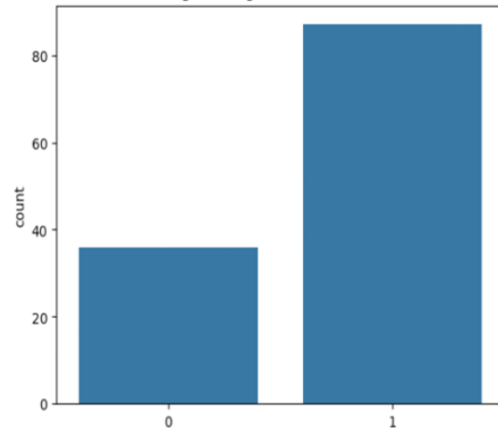
ROC-AUC Score:
0.8758513931888545

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.countplot(x=y_pred)

plt.title("Logistic Regression Predictions")

plt.show()
```

Logistic Regression Predictions



Train Random Forest model

```
rf_model=RandomForestClassifier(
    n_estimators=200,
    random_state=42
)
rf_model.fit(x_train_smote,y_train_smote)

# Random Forest Classifier
Parameters

y_pred = model.predict(x_test_processed)

# Get probabilities for ROC-AUC
y_prob = model.predict_proba(x_test_processed)[:, 1]

print("\n----- RANDOM FOREST RESULTS -----")
print("Accuracy Score:")
print(accuracy_score(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nROC-AUC Score:")
print(roc_auc_score(y_test, y_prob))
```

----- RANDOM FOREST RESULTS -----

Accuracy Score:
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Classification Report:

	precision	recall	f1-score	support
0	0.72	0.68	0.70	38
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Confusion Matrix:
[[26 12]
[10 75]]

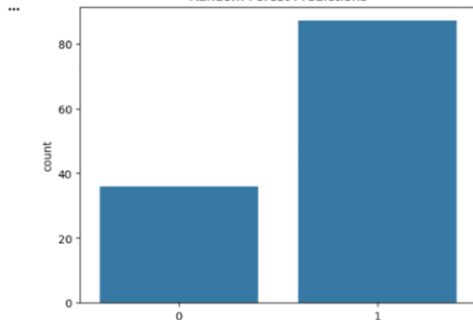
ROC-AUC Score:
0.8758513931888545

```
sns.countplot(x=y_pred)

plt.title("Random Forest Predictions")

plt.show()
```

Random Forest Predictions



Feature Importance

```
feature_names = preprocessor.get_feature_names_out()

coefficients = pd.DataFrame({
    "Feature": feature_names,
    "Coefficient": model.coef_[0]
})

coefficients = coefficients.sort_values(
    by = "Coefficient",
    ascending = False
)

print("\nTop Important Features:\n")

print(coefficients.head(10))
```

Top Important Features:

	Feature	Coefficient
4	num_Credit_History	1.264772
18	cat_Property_Area_Semiurban	0.485655
11	cat_Dependents_2	0.276412
13	cat_Education_Graduate	0.176153
0	num_ApplicantIncome	0.144487
8	cat_Married_Yes	0.143198
12	cat_Dependents_3+	0.132878
5	cat_Gender_Female	0.074236
15	cat_Self_Employed_No	0.025427
3	num_Loan_Amount_Term	-0.015256

```

import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import roc_auc_score

# Train your models
log_reg_model = LogisticRegression()
dt_model = DecisionTreeClassifier()
rf_model = RandomForestClassifier()

log_reg_model.fit(x_train_processed, y_train)
dt_model.fit(x_train_processed, y_train)
rf_model.fit(x_train_processed, y_train)

# Generate probability predictions
y_prob = log_reg_model.predict_proba(x_test_processed)[: , 1] # Logistic Regression
dt_prob = dt_model.predict_proba(x_test_processed)[: , 1] # Decision Tree
rf_prob = rf_model.predict_proba(x_test_processed)[: , 1] # Random Forest

# Calculate ROC-AUC scores
log_reg_auc = roc_auc_score(y_test, y_prob)
dt_auc = roc_auc_score(y_test, dt_prob)
rf_auc = roc_auc_score(y_test, rf_prob)

# Build results DataFrame
results = pd.DataFrame({
    "Model": ["Logistic Regression", "Decision Tree", "Random Forest"],
    "ROC-AUC": [log_reg_auc, dt_auc, rf_auc]
})

print("MODEL COMPARISON")
print(results)

```

```

... MODEL COMPARISON
      Model  ROC-AUC
0  Logistic Regression  0.852322
1      Decision Tree  0.677709
2      Random Forest  0.782353

```

```

import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
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# Generate probability predictions
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rf_prob = rf_model.predict_proba(x_test_processed)[: , 1] # Random Forest

# Calculate ROC-AUC scores
log_reg_auc = roc_auc_score(y_test, y_prob)
dt_auc = roc_auc_score(y_test, dt_prob)
rf_auc = roc_auc_score(y_test, rf_prob)

# Build results DataFrame
results = pd.DataFrame({
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    "ROC-AUC": [log_reg_auc, dt_auc, rf_auc]
})

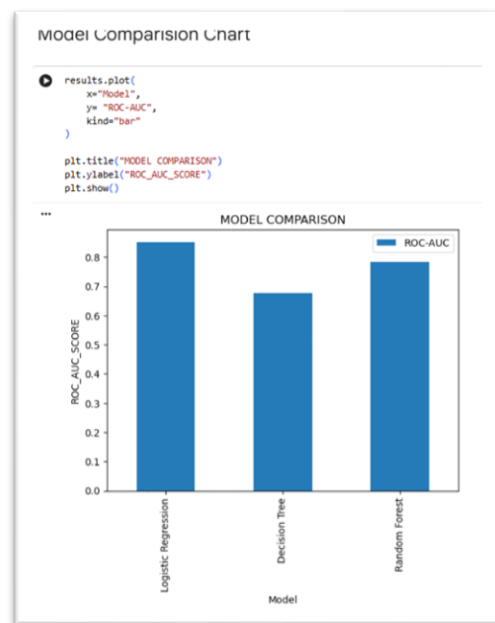
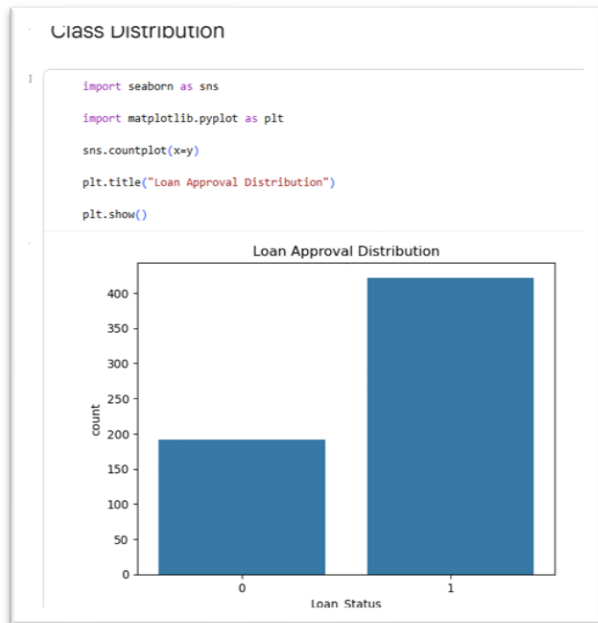
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```

```

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      Model  ROC-AUC
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```



Executive Summary :

This project aimed to develop a machine learning model for loan approval prediction using applicant demographic and financial information. Data preprocessing techniques such as handling missing values, encoding categorical variables, and feature selection were applied to prepare the dataset. Multiple classification algorithms including Logistic Regression, Decision Tree, and Random Forest were implemented and evaluated. Random Forest achieved the highest prediction performance. Feature importance analysis revealed that credit history, applicant income, and loan amount were the most significant factors affecting loan approval decisions. The project demonstrates how machine learning can assist financial institutions in making efficient, accurate, and data-driven lending decisions while reducing operational risks.